

Re-examining the Impact of FII Investments on the Indian Markets

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Abstract: This paper examines the impact of foreign institutional investors' (FIIs) investments on the Indian market over the period February 1, 2005 till January 22, 2014. The motivation for the study is that past studies on this topic are based only on examining the relationship between FII investments in cash market and any associated effects on stock market returns while ignoring the FII activity in equity derivative markets. The impact of FIIs' investments on Indian markets is re-examined in this paper by considering FII activity in Nifty index futures that trade on National Stock Exchange of India (NSE) and Singapore Exchange (SGX). The results show that Nifty index returns are statistically influenced by FII investments in Nifty index futures, although there is no causation from FIIs' cash market transactions to Nifty returns. The results indicate that index futures markets are the preferred route by FIIs for taking a position in the Indian market. Secondly, FII investments in cash market have a contemporaneous relationship with index futures activity. Thirdly, the weight of evidence shows that SGX futures are not found to be influencing Nifty, but Nifty is a significant explanatory variable in SGX futures returns.

Key words: FIIs, institutional investors, foreign portfolio flows, capital flows, Granger causality, VAR

JEL Classification: F21, F39, G15

1. Introduction:

An important development in the global financial markets from the nineties is the growth of foreign portfolio investments (FPI) in developing countries. Portfolio investments are supposed to provide benefits to both the recipient country as well as the investors, who derive the advantage of diversification, while the recipient countries will have access to the much needed capital. Although FPI helps to alleviate some structural constraints like low domestic savings in developing countries, spells of turbulences in international financial markets, such as the Asian crisis are attributed to the role played by institutional investors or at least precipitated by the actions of portfolio investors. In fact, due to their vacillating nature, FPIs also earn the sobriquet 'hot money'.

India too has welcomed FPI, almost out of compulsion, in the face of a serious balance of payments crisis. Economic reforms, which have been initiated from the early 1990s, also included opening up of the stock markets to foreign investments, have lead to a steady flow of foreign portfolio money to the equity markets. Portfolio investors in India are known as Foreign Institutional Investors (FIIs), who can invest in Indian equity as well as derivatives like stock index futures and options. Past studies (reviewed in the next section), which included research work on the influence of FPI on Indian equity market have examined

with considerable importance the relationship between FII investments and stock market returns taking into account only the cash market activity of FIIs. But FIIs can obtain exposure to Indian equity not just only in the cash market, but also through the derivative markets. Further, the SGX Nifty index futures contract that is traded on the Singapore Exchange from 2009 is also available to FIIs, which is not available for trading by the resident Indian investors. However, these aspects have not been captured in the analyses of the past studies. This study aims to examine the impact of FII investments in Indian equity markets considering FIIs activity both in the cash market and in the index futures market, thereby probably becoming the first study (at least in Indian markets) analyzing the role of FIIs both in cash and derivative markets.

The remaining part of the paper is organized as follows. Section 2 reviews the past studies; Section 3 outlines the motivation and contribution of the study; Section 4 discusses the nature and trends in FII investments in India; Section 5 describes the data; Section 6 explains the econometric methodology employed in the study; Section 7 presents the results and discussions and Section 8 provides the conclusion.

2. Literature review:

Previous studies examining the role of FIIs in the Indian stock market can be broadly classified into two categories – firstly, those works focusing on determinants of FII investments in India, and secondly, those centering around the theme of the impact of FII investments on Indian markets. Chakrabarti (2001) has examined the nature and causes of FII flows to India and observed that FII flows are highly correlated with BSE National index returns, and the index returns explain considerable variation in FII flows. The study concludes that the FII flows are “more of an effect of market returns than their cause”. Kumar (2001) has investigated the effects of FII inflows on the Indian stock market over the period 1993-1997, and reports bi-directional causality from FII flows to Sensex and vice versa. Gordon and Gupta (2003) have analyzed the factors affecting the flow of portfolio money to the Indian equity market over the period 1992-2001. Their study finds rather intriguingly a negative relationship between portfolio inflows and stock markets returns leading to the conclusion that FIIs are “not positive feedback traders but rather are bargain hunters”.

Suresh and Prabheesh (2008) have examined the interaction between FII flows and stock market returns over the period 2003-2007. They have found the existence of bi-directional causality between FII flows and stock returns, and their results indicate that FII flows are more stock return driven. Their study lends support to the information revelation hypothesis and momentum trading hypothesis.

Singh (2009) has analyzed the capital flows to India over the period spanning from 1950s to 2008, and has reported that FII inflows and stock prices have a bi-directional causal relationship with a time varying nature of the stock price volatility. His study also finds that the expansionary phase of capital inflows led to appreciation of the real exchange rate. Simultaneously, the contractionary phase of capital inflows has been associated with a depreciating real exchange rate.

Stigler, Shah and Patnaik (2010) using VAR analysis (comprising net FII investment, the Nifty index, INR/USD exchange rate, the S&P 500 index and the ADR premium index) have found that net FII flows do not cause the Nifty index returns, but the reverse causality does hold. Further, shocks to net FII flows do not carry through to the other four variables, whereas positive shocks to the exchange rate, ADR premium and S&P 500 all affect net FII flows.

Thiripalraju and Acharya (2011), using daily data over the period 2000-2009, have shown that FIIs investments are positively related to past market returns, and bi-directional causality is reported between FIIs investments and market returns. They have, therefore, concluded that over a period of time Indian mutual funds rather progressively lost their capacity to influence the market while FIIs' ability to influence has grown during the same period.

Based on post-crisis data, Bose (2012) has examined the relationship between institutional investments in India and the stock market returns. This study has found evidence of bi-directional causality between FII investments and Sensex returns, although there is no causal relation between domestic mutual funds' net investments and domestic stock returns.

Patnaik, Shah and Singh (2013) employing a modified event study methodology have focused on tail events and found that there is no unusual activity in FII investment before the day when a large positive Nifty movement took place. On the day when Nifty has an unusually large positive value, FII investment also has a statistically significant positive value. Subsequent to the event, there is evidence of slow or positive feedback trading by FIIs. Similarly, on the day when Nifty index has moved down significantly, there has been no unusual evidence of FIIs selling. With regard to the responses of Nifty index around large inflows from FIIs, they have observed significant positive returns on Nifty. Large FII outflows are not preceded by large declines in Indian stock market returns, nor trigger further negative returns.

Dua and Garg (2013) have analyzed the determinants of portfolio flows to India for the period from October 1995 to 2011, and found that stock market performance, exchange rate, domestic output growth, emerging market equity performance, interest rate differential and volatility in exchange rate are the important determinants of FII investments.

3. Motivation and contribution of the present study:

The past studies examining the relationship between FII investments and stock market returns in India have considered only the cash market investments of FIIs. But FIIs can also obtain exposure to the Indian equity through the derivative markets. However, this aspect has not been captured in the analysis of the past studies.

Derivative market provides an alternative channel through which FIIs can execute their trades first in the derivative markets followed by cash market transactions, if found necessary. As the Indian derivative markets are one of the most active markets, they can absorb transactions of considerable size without the associated price effects that typically feature the cash market transactions. In addition, transaction costs associated with derivative trading are significantly lower than the cash market transaction costs.

Apart from trading in Indian derivative markets, FIIs investment opportunity set is expanded with the availability of offshore index futures contracts based on Indian markets, especially the SGX Nifty index futures contract that is traded on the Singapore Exchange from 2009. In particular, trading in SGX Nifty futures contract offers distinct advantages to FIIs, as the transaction costs at SGX are quite low compared to those on NSE (because of the absence of the securities transaction tax and stamp duty in Singapore). The presence of alternative transmission mechanisms is not considered in the previous studies, making the conclusions of the past studies rather incomplete. Comprehensive inferences can be drawn on the relationship between FII investments and stock markets after accounting for these two variables. The present study examines the interaction between FII investments and stock market returns by broadening the independent variables to include SGX Nifty futures and FII transactions in stock index futures in addition to the cash market transactions of FIIs.

4. Nature and trends in FIIs investments in India:

1. **Stability of FII investments in the Indian Equity Market:** Starting with a modest inflow of INR 13 Crore (Cr) in 1992-1993, FII investments in equity markets during 2012-2013 peaked up to INR 140,000 Cr approximately. It can be noted that FII investments are steadily increasing during the last two decades. The average net inflows until the turn of the millennium have been INR 4955 Cr per year, while in the next five years (2000-2001 to 2005-2006), the average annual inflows have become INR 20,330 Cr. Accordingly, in the following five year period (2006-2007 to 2011-2012), the average annual inflows have increased to INR 48,492 Cr. Barring the two years following the Asian crisis and global financial crisis (1998-1999 and 2008-2009), India has been a net recipient of FII inflows.

Table 1: Trends in Net FII Investments in Indian markets in INR (Crores[#])

Financial Year	Equity	Debt	Total	Financial Year	Equity	Debt	Total
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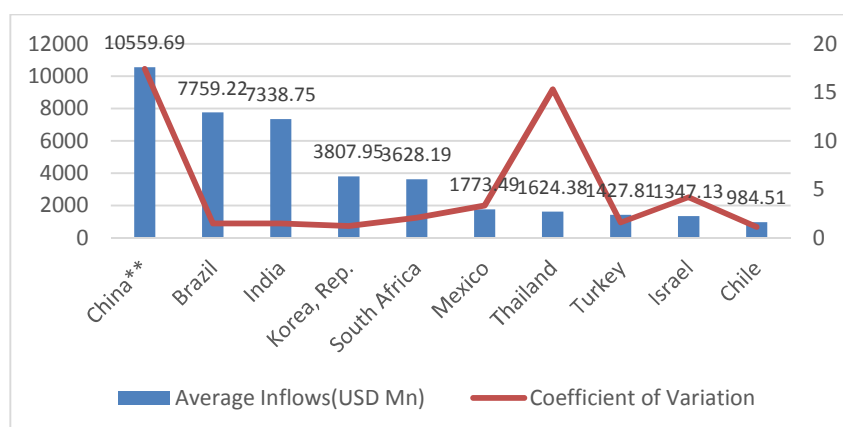
1992-93	13	0	13	2003-04	39,960	5,805	45,765
1993-94	5,127	0	5,127	2004-05	44,123	1,759	45,881
1994-95	4,796	0	4,796	2005-06	48,801	-7,334	41,467
1995-96	6,942	0	6,942	2006-07	25,236	5,605	30,840
1996-97	8,546	29	8,575	2007-08	53,404	12,775	66,179
1997-98	5,267	691	5,958	2008-09	-47,706	1,895	-45,811
1998-99	-717	-867	-1,584	2009-10	1,10,221	32,438	1,42,658
1999-00	9,670	453	10,122	2010-11	1,10,121	36,317	1,46,438
2000-01	10,207	-273	9,933	2011-12	43,738	49,988	93,726
2001-02	8,072	690	8,763	2012-13	1,40,033	28,334	1,68,367
2002-03	2,527	162	2,689	2013-14**	57,513	-63,591	-6,078

Source: SEBI Website; ** As on December 31, 2013; # one Crore = 10 millions

Figure 1 below presents the key statistics for top ten emerging markets on net Portfolio inflows over the period from 1992 to 2012. It may be observed that India is the third largest recipient of Portfolio inflows, but at the same time, the inflows are only moderately volatile as can be noted from the Coefficient of Variation (CV). In terms of the quality and quantity of inflows to India only Brazil receives larger inflows that are also relatively stable.

In the aftermath of the global financial crisis, FIIs withdrew significant amounts of money worldwide and India also experienced similar outflows, but a turnaround in the outflows can be noticed in the subsequent years. During the period from January 2008 to February 2009, FIIs withdrew INR 47,948 Cr, while in the calendar year 2008, the net outflows amount to the total of INR 53,051.70 Cr. In the subsequent months, the inflows resumed fuelled by the stimulus policy pursued in developed countries through a benevolent interest rate regime. In the month of May 2009, India alone has received INR 20,606.90 Cr., which has been attributed to the formation of a stable Central Government.

Figure 1: Net Portfolio inflows to emerging equity markets (in millions USD)

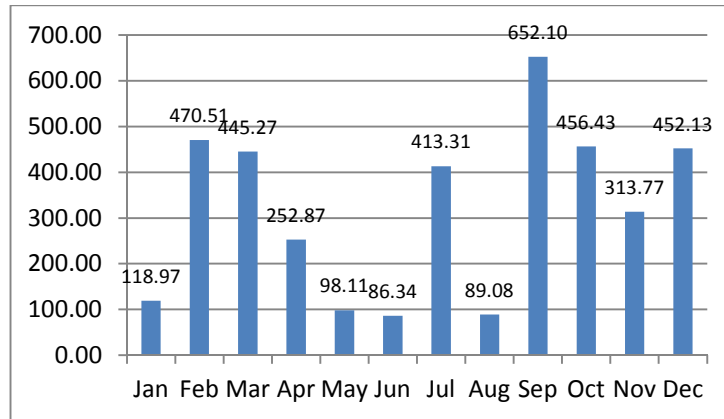


* Source: World Bank (prepared from the World Development Indicators (WDI).
<http://data.worldbank.org/indicator/BX.PEF.TOTL.CD.WD/countries/1W-IN?display=default>

**data available from 1997;

2. **Seasonality in FII investments to India:** Figure 2 presents the calendar month-wise average net FII inflows during the period 2000-2014. FII inflows are significantly higher during the last four months of the calendar year compared to the other months.

Figure 2: Average monthly net flows (INR Cr) from February 2000 to January 2014



Source: Compiled from SEBI's web site

To examine the presence of any seasonality patterns, monthly net inflows have been regressed on monthly dummies. To detect monthly seasonality, the following regression is estimated (Brooks, 2008):

$$\text{net FII}_t = \gamma_1 D1_t + \gamma_2 D2_t + \gamma_3 D3_t + \gamma_4 D4_t + \gamma_5 D5_t + \gamma_6 D6_t + \gamma_7 D7_t + \gamma_8 D8_t + \gamma_9 D9_t + \gamma_{10} D10_t + \gamma_{11} D11_t + \gamma_{12} D12_t + u_t$$

where, D1 = January, D2 = February ... D12 = December.

The regression coefficient can be interpreted as the average net FII inflow for that particular given month. Similar methodology has been used by Brooks and Persaud (2001) to examine the day of the week effect in Southeast Asian Markets.

The regression results presented in Table 2 below indicates that inflows in the months of February and March have been significantly higher, and is generally associated with the pre and post budget rallies. The inflows are statistically significant during the month of July probably due to the resolution of the uncertainty regarding monsoon. Again, the inflows are significant in September to December period, which could be due to the anticipated market movements associated with half yearly results and the festive periods. Indian stock market have experienced pronounced seasonality effect (month of year effect) as has been reported by Chakrabarti and Sen (2007) and Parikh (2009), and hence, the stock market rallies are observed during these periods.

Table 2: Dummy variable regression to examine seasonality (dependent variable: net FII inflows)

Variable	Coefficient	t-ratio	Variable	Coefficient	t-ratio
D1	1427.7	0.6986	D7	4959.67	2.427**
D2	5646.16	2.7629***	D8	1068.97	0.5231

D3	5343.19	2.6147***	D9	7825.24	3.8293***
D4	3034.4	1.4849	D10	5477.2	2.6803***
D5	1177.28	0.5761	D11	3765.22	1.8425*
D6	1036.06	0.507	D12	5425.52	2.655***

*10% level of significance; **5% level of significance; 1% level of significance;

D_i dummy variable for the month i = 1 (Jan) to 12 (Dec)

3. **Preferred sectors of FIIs:** FIIs play an important role in Indian markets, as liquidity providers and the average turnover of the FIIs in the equity market constitute around 16.0 percent of the total turnover on the BSE and the NSE during 2006-2007 to 2012–2013. Apart from the Indian Promoters and the individual investors, FIIs have emerged as a significant shareholding category of investors in Indian companies (listed on NSE). Banking and FMCG sectors have attracted considerable interest from the FIIs. However, the bulk of the FII investments is going into large companies and small companies are not getting the benefit of FII's money (Department of Economic Affairs, Ministry of Finance, 2010).
4. **FIIs activity in equity derivative markets:** FIIs registered with SEBI are permitted to trade exchange-traded derivatives, and are subjected to position limits as specified by SEBI. Table 3 shows the FII activity on Futures and Options (F & O) segment of NSE, these are aggregate numbers, including contracts on index futures, index options, single stock futures and single stock options. It may be observed that FIIs are net buyers in each of the years from 2006 to 2013. Even in the year 2008-2009, when FIIs are net sellers in the cash markets, they are net buyers in the F & O segment of the NSE. Analysis of the daily activity of FIIs in the index futures market from January 2010 to January 2014 shows that almost 43% of the time FIIs' cash market and derivative markets' activities are opposite in direction, i.e., If they are net buyers (sellers) in cash market, they are net sellers (buyers) in the index futures market. While 37% of the time they are net buyers in both cash market and index futures market, and in the remaining 20% of the time, they are simultaneously net sellers in both the markets. Daily average (5% trimmed mean) purchases of FIIs in the index futures market is INR 2355.57 Cr and the average sales amount to the total of INR 2366.77 Cr.

Table 3: FII activity in Futures & Options segment on NSE

Year	Buy side amount in mn	Sell side amount in mn	FII turnover to total F&O turnover on NSE (in %)
2006-07	Rs. 64,84,665	Rs. 63,60,392	8.7
2007-08	Rs. 1,26,33,510	Rs. 1,24,01,616	9.6
2008-09	Rs. 1,01,65,351	Rs. 96,80,224	9

2009-10	Rs. 1,75,19,697	Rs. 1,72,52,480	9.8
2010-11	Rs. 3,18,53,782	Rs. 3,05,18,144	10.7
2011-12	Rs. 4,57,15,596	Rs. 4,48,92,230	14.5
2012-13	Rs. 4,95,58,023	Rs. 4,87,74,962	15.6

Source: Indian Securities Market Review 2013, NSE

5. Data description and definition of variables:

The data used in this study comes from different sources - daily FIIs' activity on the cash market and futures markets is obtained from SEBI's website, and the Nifty index data and the SGX Nifty are obtained from their respective websites. In this study, only FII activity in Nifty index futures is considered because of two reasons – 1) A competing contract SGX Nifty futures is available in the overseas market (Singapore Exchange) with considerable volumes and associated with lower transaction costs, and 2) Trading in index futures is a clear and economically efficient manner to bet on the market direction that can be suitably used for speculation as well as hedging purposes. The study spans the period [1] from February 1, 2005 till January 22, 2014 using daily observations on the following four variables:

Net FIIs' investments in cash market (NetFF): Following Sellin (1996), the difference in the daily cash market purchases of FIIs (P_c) and cash market sales of FIIs (S_c) is defined as net FIIs' investment in the cash market. It is normalized by dividing it with the total FII turnover, i.e., $NetFF = \frac{P_c - S_c}{P_c + S_c}$

Net FIIs' investments in the index futures market (NInF): The difference in the daily purchases of FIIs in Nifty index futures (P_d) and Nifty index futures sales of FIIs (S_d) is defined as net FIIs' activity in the futures market. It is normalized by dividing it with the total FII turnover, i.e., $NInF = \frac{P_d - S_d}{P_d + S_d}$

Nifty Index returns (Nifty): Nifty index is the benchmark index of the National Stock Exchange (NSE) India. The index returns are computed as the natural log differences, i.e., $Nifty = \ln(I_t/I_{t-1})$, where, I_t is the level of the Nifty index on day 't'.

SGX futures returns (SGX): SGX is the future contract trading on Singapore Stock Exchange with Nifty index as the underlying asset. The returns are computed as the natural log differences, i.e., $SGX = \ln(F_t/F_{t-1})$, where, F_t is the future level on day 't'.

6. Methodology:

In this study, Vector Auto Regression (VAR) methodology has been used to analyze the effect of FII investments in Indian capital markets. VAR is a standard tool used by Froot *et al* (2001), Dahlquist and Robertson (2004) and Richards (2004) to examine the relationship between net foreign inflows and market returns. VAR is a dynamic system of equations in which the current level of each variable in the

system depends on past movements in that variable and in all the other variables in the system. A VAR system can be expressed in the following form:

$$Y_t = A_1 Y_{t-1} + \dots + A_p Y_{t-p} + \varepsilon_t$$

where, Y_t is a vector of endogenous variables at time t and $A_i = (i = 1 \dots p)$ are coefficient vectors, p is the number of lags included in the system, and ε_t is a vector of residuals. The appropriate lag lengths are selected using the Akaike Information Criteria (AIC).

In order to ascertain the causality between the FII activity and market returns, Granger Causality/ Block Exogeneity Wald test has been applied. If a variable, or a block of variables, are strictly exogenous, then the implied zero restrictions ensure that these variables do not react to a shock to any of the endogenous variables inferring that the lags of one variable can Granger-cause any other variables in the VAR system. The null hypothesis is that all lags of one variable can be excluded from a given equation in the VAR system. For instance, the rejection of the null from the Nifty equation means that all lags of NetFF cannot be excluded, and there is causality between NetFF and Nifty. Asymptotically, the test statistic given below is χ^2 distributed with degrees of freedom equaling the number of restrictions in the system (Enders, 1995).

$$(T - c) (\log |\Sigma_r| - \log |\Sigma_u|)$$

where, T is the number of observations; Σ_u is variance/covariance matrices of the unrestricted VAR system; Σ_r is variance/covariance matrices of the restricted system when the lag of a variable is excluded from the VAR system; c is the maximum number of regressors in the longest equation.

Following Sims' (1980) seminal work, dynamic analysis of VAR model is complemented by examining the VAR impulse responses and variance decompositions because Granger Causality/ Block Exogeneity Tests will suggest the statistical significance of the variables that have significant impact on the future values only, but these tests do not reflect interactions between the variables in the VAR system. To address this issue, impulse-response function and the variance decompositions have been employed. The impulse response function traces the responsiveness of dependent variables in the VAR to shocks from all the explanatory variables. If the system of equations is stable, any shock should decline to zero. The impulse response function is defined as:

$$IR(m, h, Z_{t-1}) = E(y_{t+m} | e_t = h, Z_{t-1}) - E(y_{t+m} | Z_{t-1})$$

where, m denotes the time, $h = (h_1, \dots, h_m)'$ is $n \times 1$ vector denoting the size of the shock, Z_{t-1} denotes cumulative information from the past up to time $t-1$ (Pesaran and Shin, 1998). For calculating impulse responses and variance decompositions, the ordering of the variables is important. If the error terms are correlated, the results will be sensitive to the ordering of the variables. In this study, the generalized

impulse response analysis has been carried out, which is recently proposed by Pesaran and Shin (1998), as it is invariant to the ordering of the variables in the VAR. In contrast to the traditional orthogonalized impulse responses computed using the Choleski decomposition method; the generalized impulse response functions are unique.

While impulse response functions (IRF) trace the effects of a shock to one endogenous variable on to the other variables in the VAR, variance decomposition separates the variation in an endogenous variable into the component shocks to the VAR. Hence, the study also examines the variance decompositions. Forecast error variance decomposition (FEVD) is an econometric tool used in the vector auto-regression context for assessing the driving forces of the dependent variables. In particular, an examination of the variance decompositions indicates the percentage of the forecast error variance in one variable, that is, due to errors in forecasting itself and each of the other variables.

7. Results and Discussions

The analysis is conducted by dividing the data into two sub-periods – the first sub-period is from February 1, 2005 to November 18, 2009 and the second period is from November 20, 2009 to January 22, 2014. The second sub-period coincides with the availability of SGX Nifty contract. Prior to estimating the VAR equations, it is mandatory to check that each univariate series is nonstationary or $I(1)$. Some commonly used unit root tests are the augmented Dickey-Fuller test (ADF test), the Phillips-Perron test (PP test) and the Kwiatkowski-Phillips-Schmidt and Shin test (KPSS test), which have been carried out to investigate the unit roots. As of now there is still no consensus on any one particular test as the most powerful unit root test, generally ADF and PP have now been accepted in the literature as primary tests for unit roots and KPSS test can be used to confirm results from the ADF and PP tests.

A problem common with the conventional unit root tests, such as the ADF, DF-GLS and PP-tests, is that they do not allow the possibility of a structural break. Zivot and Andrews (1992) have proposed a unit root test accounting for structural change in which data dependent algorithm is used to determine the break points. In this study, Zivot and Andrews (1992) test that combines one-time changes in the level and the slope of the trend function of the series has been employed. The null hypothesis is a unit root with a structural break in intercept as well as trend. In the ADF tests, the null hypothesis is that a time series Y_t is $I(1)$, and the alternative hypothesis is that it is $I(0)$, while in the KPSS test, the null hypothesis of stationarity is tested.

Table 4: Unit Root Test results

Feb 2005 to Jan 2014				
Variable	ADF test statistic	PP test statistic	KPSS test statistic	ZA test statistic

Nifty	-44.5701	-44.5071	0.1266	-44.7653
NetFF	-12.54709	-35.8058	0.3067	-13.5211
NInF	-40.9642	-40.8862	0.289	-23.3496
1% critical values	-3.4331	-3.4311	0.7390	-5.57
Nov 2009 to Jan 2014				
SGX	-32.7533	-32.7617	0.0426	-32.8434
1% critical values	-3.4364	-3.4364	0.7390	-5.57

*The lag length is determined based on AIC;

All the tests reject the presence of a unit root for all the four variables used in the study. So, all the four variables used in the study are stationary, and fulfill the requirements of the VAR model.

Next, VAR equations are estimated with a lag length of three, based on Akaike information criteria. The VAR analysis is conducted by dividing the sample period in to two parts – the first period spans February 2005 to November 2009 and the second period starts from November 2009 to January 2014. The basis for the division of the sample period is the availability of SGX Nifty futures contract for trading on Singapore Exchange. The results of VAR during the first period (Feb 2005 to Nov 2009) are reported in Table 5.

Table 5: Vector auto regressions of the system of Nifty, NInF and NetFF (Feb 2005 to Nov 2009)

Variable	Nifty	NetFF	NInF	Variable	Nifty	NetFF	NInF
Nifty(-1)	0.011446	1.199212	0.759340	NetFF(-3)	0.004853	0.147410	-0.04704
t-statistic	0.32933	5.55778	1.7387	t-statistic	1.00084	4.89643	-0.77196
Nifty(-2)	-0.07285	-0.2552	-0.67912	NInF(-1)	0.004511	0.024824	0.071636
t-statistic	-2.07403	-1.17041	-1.5388	t-statistic	1.70841	1.51436	2.15909
Nifty(-3)	-0.03173	-0.03617	-0.88486	NInF(-2)	0.004471	0.007192	0.032075
t-statistic	-0.90743	-0.16657	-2.01364	t-statistic	1.69556	0.43935	0.96808
NetFF(-1)	0.005460	0.268418	-0.04084	NInF(-3)	0.001495	0.002723	0.080474
t-statistic	1.07634	8.5235	-0.64074	t-statistic	0.57135	0.16761	2.44697
NetFF(-2)	0.000266	0.136522	-0.06887	C	0.000285	0.015380	0.016398
t-statistic	0.05158	4.26785	-1.06377	t-statistic	0.46203	4.0153	2.1151
R-squared	0.012296	0.258221	0.027944	F-statistic	1.608668	44.98359	3.714746

Note: NetFF = Net FIIs' investments in cash market; NInF = Net FIIs' investments in index futures market; Nifty = Nifty Index returns; Figures in parenthesis indicate the number of lags.

As VAR modelling is a-theoretical in nature and given the large number of parameters involved along with the interconnectivity of the equations, the estimated coefficients are rather difficult to interpret (Brooks,

2008). Hence, Granger Causality Tests, IRF and FEVD have been employed. The results of Granger Causality tests for the data spanning from February 1, 2005 to November 18, 2009 have been presented in Table 6.

Table 6: Granger Causality/Block Exogeneity Wald Tests Feb 2005 to Nov 2009)

Null Hypothesis	Chi. Sqr	P-value	Inference
NetFF does not Granger causes Nifty	3.6109	0.3067	do not reject
NInF does not Granger causes Nifty	6.5703	0.0869	Reject
Nifty does not Granger causes NetFF	33.3517	0.0000	Reject
NInF does not Granger causes NetFF	2.6037	0.4568	do not reject
Nifty does not Granger causes NInF	9.7831	0.0205	Reject
NetFF does not Granger causes NInF	4.5425	0.2085	do not reject

Marginal significance levels associated with Chi Square tests;

Granger Causality Test Results: Granger Causality test is a simple test to ascertain whether a particular variable is affected by past values of other variables. An important advantage of this test is that it is unaffected by the ordering of the variables in the VAR system. The results (Table 6) indicate that there is bi-directional causality from NInF to Nifty. It can also be noted that Nifty Granger causes NetFF. Therefore, there is feedback between FII's activity in the futures market and the Nifty index. But only unidirectional causality from Nifty to NetFF which implies that Nifty changes precede changes in NetFF.

Impulse Response Functions: An impulse response refers to the reaction of any dynamic system in response to some external change. The graph [2] of the impulse response function is shown in Figure 3. The top panel shows the response of NetFF, the next panel shows the impulse response of Nifty, and the final panel pertains to that of NInF. The graphs depict the response of each variable to a shock in itself as well as to shocks in the remaining variables. The dotted lines around the thick middle line (impulse responses) indicate the two standard deviation bands. From the graph of IRF, it is evident that shocks to NetFF die down in around ten days. The response of Nifty to shocks in either NetFF or NInF is similar in size and pattern, and they are dissipated in around four days. A one standard deviation shock originating from Nifty elicits a greater response from NInF vis-à-vis shocks from NetFF. However, the shocks are absorbed by NInF in around five days.

Figure 3: Generalized Impulse Response functions (Feb 2005 to Nov 2009)

Response to Generalized One S.D. Innovations ± 2 S.E.

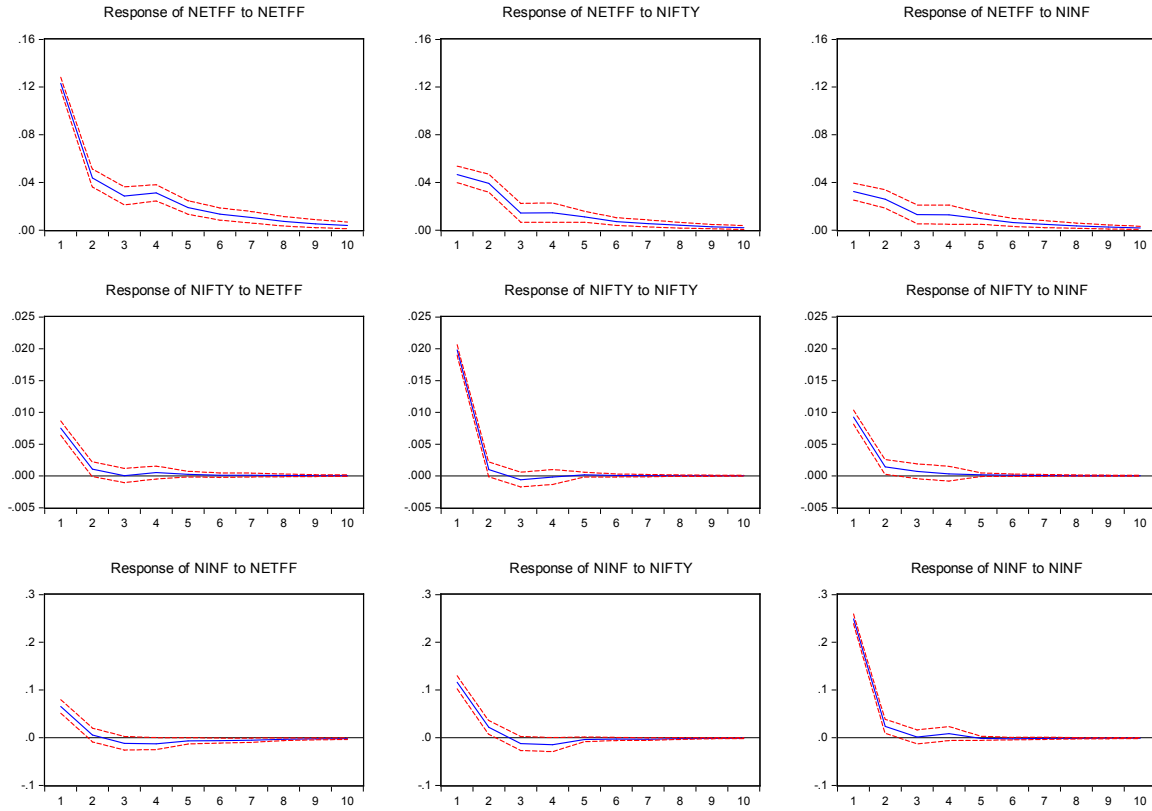


Table 7: Ten day Forecast error variance decomposition (in percentage)

Variance Decomposition of NetFF:				Variance Decomposition of Nifty:			
Period	NetFF	Nifty	NInF	Period	NetFF	Nifty	NInF
1	84.69575	8.423136	6.881118	1	0.000000	78.22703	21.77297
2	77.78257	12.51637	9.701067	2	0.096690	77.76911	22.13420
3	77.40036	12.41981	10.17983	3	0.104662	77.72406	22.17127
4	77.22661	12.26802	10.50537	4	0.190958	77.64350	22.16554
5	76.92396	12.32042	10.75562	5	0.199672	77.63080	22.16953
6	76.82783	12.31625	10.85593	6	0.201309	77.62732	22.17137
7	76.77811	12.30914	10.91275	7	0.203881	77.62434	22.17178
8	76.74573	12.30898	10.94529	8	0.204828	77.62328	22.17189
9	76.73072	12.30806	10.96122	9	0.205197	77.62286	22.17194
10	76.72286	12.30739	10.96975	10	0.205443	77.62261	22.17195
Variance Decomposition of NInF:							
Period	NetFF	Nifty	NInF	Period	NetFF	Nifty	NInF
1	0.000000	0.000000	100.0000	6	0.398396	1.293754	98.30785
2	0.034090	0.223911	99.74200	7	0.433169	1.306289	98.26054
3	0.159670	0.554839	99.28549	8	0.447061	1.309500	98.24344
4	0.299247	1.268866	98.43189	9	0.455277	1.310824	98.23390
5	0.351309	1.286641	98.36205	10	0.460140	1.311744	98.22812

Note: NetFF = Net FII's investments in cash market; NInF = Net FII's investments in index futures market; Nifty = Nifty Index returns;

Forecast Error Variance Decomposition: It is clear from FEVD (Table 7) that NInF explains a fifth of the variance in Nifty while NetFF variations have an extremely limited role in explaining the variance in Nifty. Although causality tests do not indicate causality from NInF to NetFF, variance decomposition of NetFF indicates that NInF and Nifty together explain a large part of the variance in NetFF. FEVD of NInF shows that variations in Nifty explain only a small extent of the variance in NInF. This indicates that NInF is least influenced by innovations in either Nifty or NetFF, at the same time, it is a dominating variable in explaining variations in Nifty as well as in NetFF.

VAR equations during the period from November 20, 2009 to January 22, 2014 are estimated with a lag length of six (indicated by Akaike information criteria) and the VAR dynamics are presented in Table 8 followed by Granger Causality test results in Table 9.

Table 8: Vector auto regressions of the system of Nifty, SGX, NInF and NetFF (Nov 2009 to Jan 2014)

	NInF	NetFF	Nifty	SGX		NInF	NetFF	Nifty	SGX
NInF(-1)	0.180858	0.026992	0.005281	0.006243	Nifty(-2)	-4.80596	-4.59044	-0.37604	0.082737
t-statistic	4.98989	1.10936	2.08218	2.30587	t-statistic	-2.06873	-2.94351	-2.31315	0.47679
NInF(-2)	0.062438	0.022209	0.001927	0.001300	Nifty(-3)	-7.83542	-5.12188	-0.5063	-0.14491
t-statistic	1.7108	0.90652	0.75456	0.47699	t-statistic	-3.21548	-3.13111	-2.9692	-0.79613
NInF(-3)	-0.00853	0.011430	0.007974	0.008220	Nifty(-4)	-8.26699	-6.09812	-0.35668	-0.13191
t-statistic	-0.23486	0.4688	3.13743	3.02995	t-statistic	-3.39887	-3.73482	-2.09561	-0.72605
NInF(-4)	0.039144	0.016752	-0.00051	-0.00051	Nifty(-5)	-2.34133	-5.62579	-0.27551	-0.15726
t-statistic	1.07187	0.68336	-0.19814	-0.18501	t-statistic	-1.01551	-3.6349	-1.70767	-0.91311
NInF(-5)	-0.06226	-0.00172	-0.00188	-0.00226	Nifty(-6)	-0.93081	-3.04811	-0.12814	-0.1223
t-statistic	-1.707	-0.07025	-0.73758	-0.83016	t-statistic	-0.48158	-2.34925	-0.9474	-0.8471
NInF(-6)	0.045890	0.008418	0.000839	0.000774	SGX(-1)	6.314886	5.626095	0.203177	-0.48545
t-statistic	1.2862	0.35147	0.33615	0.29039	t-statistic	3.47926	4.61758	1.59971	-3.58067
NetFF(-1)	0.038681	0.303504	-0.00259	-0.00408	SGX(-2)	4.162383	4.647100	0.325653	-0.14083
t-statistic	0.72135	8.43135	-0.68971	-1.01802	t-statistic	1.88133	3.1289	2.10342	-0.85216
NetFF(-2)	0.034926	0.161958	0.006632	0.008728	SGX(-3)	7.047211	5.225480	0.379518	0.017852
t-statistic	0.62439	4.3132	1.6943	2.08885	t-statistic	3.02089	3.3368	2.32486	0.10245
NetFF(-3)	-0.04092	-0.00325	-0.00519	-0.00655	SGX(-4)	8.058261	5.962847	0.349728	0.137980
t-statistic	-0.72847	-0.08618	-1.31963	-1.5607	t-statistic	3.45537	3.80884	2.14304	0.79208
NetFF(-4)	-0.0494	0.109276	0.000738	0.000959	SGX(-5)	3.510704	5.228325	0.285999	0.183017
t-statistic	-0.88005	2.89974	0.18784	0.22878	t-statistic	1.59124	3.53012	1.85248	1.11053
NetFF(-5)	-0.10609	0.081280	-0.00032	-0.00107	SGX(-6)	0.984550	2.474444	0.150156	0.158768
t-statistic	-1.90405	2.17314	-0.08217	-0.25741	t-statistic	0.54231	2.03038	1.18196	1.17077
NetFF(-6)	0.032567	0.073777	-0.00028	-0.00061	C	0.002890	0.017598	0.000381	0.000489
t-statistic	0.63745	2.15114	-0.07864	-0.16016	t-statistic	0.4771	4.32835	0.89991	1.08159

Nifty(-1)	-5.88137	-3.55268	-0.19331	0.488468	R-squared	0.097454	0.421740	0.039033	0.046431
t-statistic	-3.00052	-2.69998	-1.40935	3.3362	F-statistic	4.557522	30.78360	1.714442	2.055178

Note: NetFF = Net FIIs' investments in cash market; NInF = Net FIIs' investments in index futures market; Nifty = Nifty Index returns; SGX = SGX futures returns. Figures in parenthesis indicate the lag.

Table 9: Granger Causality/Block Exogeneity Wald Tests

Null Hypothesis	Chi sqr	P-value	Inference
NetFF does not Granger causes Nifty	3.9694	0.6808	Do not reject
SGX does not Granger causes Nifty	7.1257	0.3094	Do not reject
NInF does not Granger causes Nifty	17.5161	0.0076	Reject
Nifty does not Granger causes NetFF	20.6377	0.0021	Reject
SGX does not Granger causes NetFF	30.2468	0.0000	Reject
NInF does not Granger causes NetFF	3.5850	0.7326	Do not reject
Nifty does not Granger causes NInF	19.5371	0.0033	Reject
NetFF does not Granger causes NInF	9.0141	0.1728	Do not reject
SGX does not Granger causes NInF	21.7019	0.0014	Reject

Marginal significance levels associated with Chi Square tests;

Note: NetFF = Net FIIs' investments in cash market; NInF = Net FIIs' investments in index futures market Nifty = Nifty Index returns; SGX = SGX futures returns

Granger Causality Test Results: Bi-directional causality is indicated between FII activity in the index futures and Nifty returns. Therefore, there is feedback between the Nifty movements and FIIs' investments in index futures. This finding is consistent with that observed in the previous period (Feb 2005 to Nov 2009). Granger Causality is observed from the Nifty index to the cash market activity of FIIs. This is akin to the findings in the first sub-period as well as that reported in the past studies on Indian market. However, there is no causality in the reverse direction, i.e., from NetFF to Nifty, which is in contrast to the findings from earlier studies. Unidirectional causality is implied from SGX to NetFF and finally, there is no causality from NInF to NetFF.

Impulse Response Functions: IRF graph (Figure 4) shows that one standard deviation disturbance in Nifty or SGX leads to a similar response from NInF. A similar response is elicited by Nifty and SGX from NetFF too, and the shocks persist for around ten days before returning to equilibrium. A one standard deviation disturbance originating in SGX results in an approximately 0.1 percent increase in Nifty returns and the

price adjustment are fairly quick, whereas, innovations in NetFF and NInF do not seem to elicit a similar response from Nifty.

Figure 4: Generalized Impulse Response functions

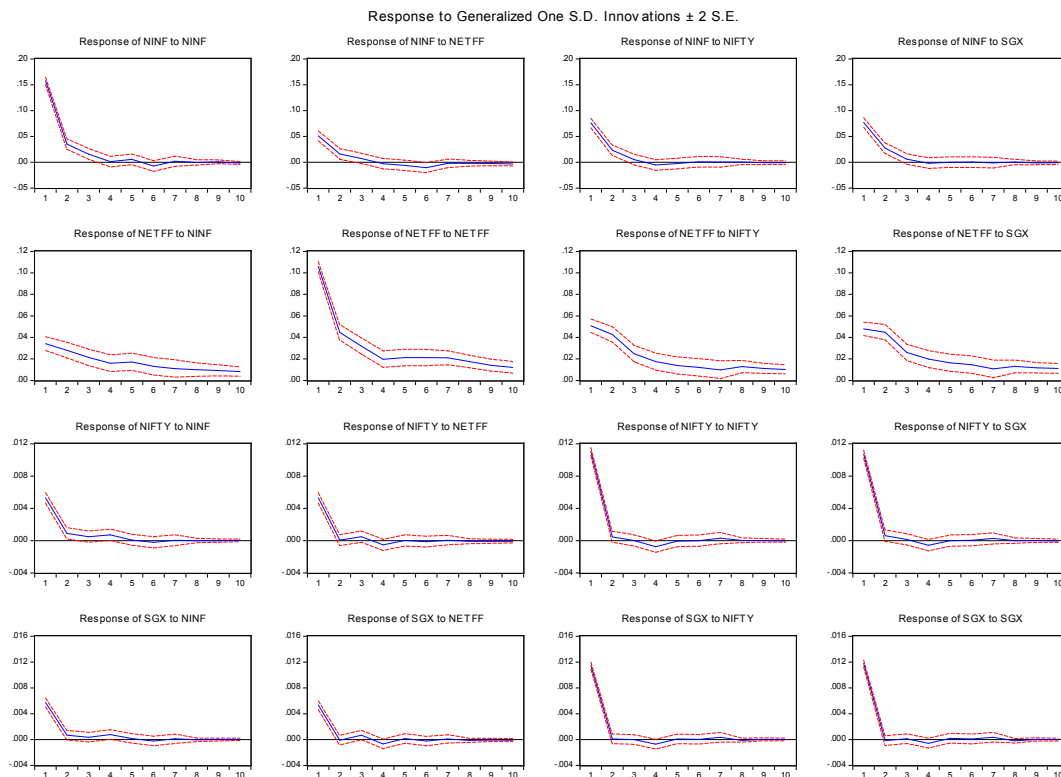


Figure 4 above shows Generalized Impulse Response functions. Responses are expressed in standard deviation units. The time scale on the horizontal axis is expressed in days. Results are based on the vector auto regression (VAR) specified in Table 8. For each impulse response function $\pm 2SE$ (dotted lines) are also reported.

Forecast Error Variance Decomposition: Variance decomposition estimated in Table 10 suggests that the disturbance originating from NInF has inflicted the greatest variability to NInF and the proportion of variance stays high even at the end of the tenth day. Reasonably high proportion of variance in Nifty returns is explained by NInF together with NetFF, as they account for more than 30% of the variance in Nifty. In case of fluctuations in NetFF, they are mainly explained by variance in NInF along with NetFF, and the proportion increases gradually from around 10% to around 18.75% by the end of the tenth day. Around 60% of the variance in SGX is explained by Nifty and another 23% is explained by NInF. In a nutshell, NInF shocks are an important source of fluctuations to NetFF, NInF and SGX movements. But FII activity in cash market has only a limited impact on Nifty index as well as on SGX.

Table 10: Ten day Forecast error variance decomposition (in percentage)

Variance Decomposition of NInF:					Variance Decomposition of NetFF:				
Period	NInF	NetFF	Nifty	SGX	Period	NInF	NetFF	Nifty	SGX

1	100.0000	0.000000	0.000000	0.000000	1	10.38568	89.61432	0.000000	0.000000
2	98.66227	0.087034	0.128445	1.122246	2	13.87679	81.54062	2.906735	1.675856
3	98.54579	0.104138	0.195354	1.154720	3	15.68872	79.33011	2.986070	1.995099
4	97.66491	0.145311	0.326848	1.862932	4	16.62014	77.47273	3.051883	2.855247
5	97.00456	0.402707	0.359379	2.233358	5	17.66390	75.71650	2.918437	3.701169
6	96.43284	0.672029	0.656337	2.238793	6	17.98588	74.73785	2.810183	4.466085
7	96.26928	0.704576	0.655734	2.370411	7	18.15782	74.54877	2.758767	4.534636
8	96.23316	0.722996	0.666723	2.377120	8	18.36226	74.36876	2.774199	4.494778
9	96.19536	0.753265	0.667526	2.383849	9	18.57459	74.10167	2.794202	4.529535
10	96.16001	0.786698	0.670447	2.382844	10	18.72699	73.81481	2.829560	4.628644
Variance Decomposition of Nifty:					Variance Decomposition of SGX:				
Period	NInF	NetFF	Nifty	SGX	Period	NInF	NetFF	Nifty	SGX
1	23.15338	11.77496	65.07166	0.000000	1	23.74139	9.719917	61.17680	5.361884
2	23.56859	11.71162	64.46998	0.249807	2	23.65442	9.643434	60.18524	6.516901
3	23.59150	11.72693	64.17708	0.504487	3	23.57029	9.821371	59.90653	6.701814
4	23.52923	12.01140	63.73824	0.721132	4	23.53313	10.32352	59.44008	6.703267
5	23.51382	12.00133	63.69787	0.786982	5	23.51018	10.31461	59.34293	6.832273
6	23.53351	11.99731	63.66746	0.801728	6	23.52029	10.32336	59.30678	6.849580
7	23.51264	11.98692	63.69002	0.810422	7	23.50621	10.31605	59.33407	6.843674
8	23.50277	11.98968	63.66738	0.840175	8	23.48412	10.32011	59.27596	6.919806
9	23.50003	11.99482	63.66378	0.841368	9	23.48282	10.32267	59.27432	6.920191
10	23.49864	12.00147	63.65859	0.841300	10	23.48048	10.33235	59.26756	6.919611

Note: NetFF = Net FIIs' investments in cash market; NInF = Net FIIs' investments in index futures market; Nifty = Nifty Index returns; SGX = SGX futures returns

From the analysis, it is emerging that FIIs first take positions in SGX contract that trades on the Singapore exchange, which is open even while Indian markets are closed due to time zone differences. Subsequently, as the Indian market opens up FIIs might be taking positions initially in the index futures markets, which are later converted into cash market positions. The actions of FIIs in Nifty index futures contract and in SGX contract lead the movements in the Nifty index, hence, causality is observed between FIIs activity in the futures market and Nifty.

To explore whether there is any contemporaneous relationship between NetFF and NInF, OLS regression is performed with NetFF as the dependent variable. The results are presented in the Table below, which indicate that there is a contemporaneous relationship between FIIs' investments in cash market and in the index futures market.

Table 11: Contemporaneous regression between FIIs activity in cash and index futures market (Feb 2005 to Jan 2014)

	Coefficient	t-ratio
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Const	0.0493	16.9445***
NInF	0.1559	11.5079***

*** 1% level of significance

On the basis of the above econometric analysis, it may be inferred that FIIs are increasingly using index futures markets to express their views on the Indian market before taking positions in the cash market probably because of the lower transaction costs, depth of the futures market and also may be for avoiding front-running. Nifty index movements trigger FII investments both in the cash market as well as index futures markets. However, only the FII activity in index futures sets off Nifty index movements leading to the conclusion of positive feedback trading. In general, the policymakers are concerned whenever the FIIs are significantly influencing the local market movements. These concerns are grounded on the premise that FIIs may exhibit *irrational exuberance* both in the built up of equity market bubbles as well as during any sell offs even though they may not lead them. While these concerns are not misplaced, the bi-directional causality observed in the Indian markets is supposed to not elicit serious apprehensions, since the feedback is observed in the index futures markets. In addition, the IRF graphs indicate that the impact of cash market activity of FIIs has a nominal impact on the Nifty index, and it is their activity in the futures market, which has considerable impact, as the equity derivative markets in India have considerable depth and liquidity with considerable volumes attributed to non-FII investors. Further, the presence of overseas index futures markets will also be helpful in absorbing a part of the trades during any sell offs. Policymakers and regulators now may have to watch the FII activity in the derivatives market more than their cash market activity. However, given the causality between FIIs activity in index futures and Nifty in conjunction with the contemporaneous relationship between FIIs activity in futures and cash market activity, any spurts of withdrawals may still have the potential to weaken the equity markets. Therefore, the current restrictions on FIIs positions in equity derivative markets ought to continue, so that FIIs use the derivative markets only for hedging purposes.

8. Conclusions:

This study examines the interaction between FII activity in both cash market, index futures market and the stock market returns. This study considers index futures of the trades in India as well as the trades in SGX contract. Indian equity markets attract steady inflows from FIIs ever since the markets have been opened in 1992. Excluding the post Asian crisis and global financial crisis in 2008, India is a net recipient of FII inflows. FII investments are significantly higher in the last four months of a calendar year, and also during the pre and post budget period. Using VAR framework and Granger Causality tests, the study finds that Nifty index returns are statistically influenced by FII investments in Nifty index futures, although there

is no causation from FIIs' cash market transactions to Nifty returns. The results indicate that index futures markets are the preferred route by FIIs for taking a position in the Indian market. Therefore, for investors, FIIs activity in the index futures market may be more informative than their activity in the cash market for predicting the direction of Indian markets. Secondly, Net FII investments in cash market have a contemporaneous relationship with index futures activity. Thirdly, the weight of evidence shows that SGX futures are not found to be influencing Nifty, but Nifty is a significant explanatory variable in SGX futures returns. In conclusion, regulators may have to continue with the current restrictions on FIIs positions in futures markets and the derivative markets be used by FIIs for hedging purposes only.

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End Notes:

- [1] The starting point of the study is motivated by the earliest point of time from where data is available on a continuous basis on the Web site of SEBI.
- [2] Only Impulse response graphs are presented in this paper. The Table of coefficients can be obtained from the authors upon request.